INCH-POUND

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22 November 1988
SUPERSEDING
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3 December 1987 and
MIL-S-52868B
27 June 1979

MILITARY SPECIFICATION

SENSOR, CAPACITANCE PROXIMITY

DT-548/FSS-9 (V)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a capacitance proximity sensor designed to detect the presence of an intruder in close proximity to or in contact with protected metal objects. It is a component of the joint-services interior intrusion detection system (J-SIIDS).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated- Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Set-up.

MILITARY

MIL-P-116	- Preservation, Methods of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-781	- Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution.
MIL-STD-810 1/	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods.
MIL-STD-45662	- Calibration System Requirements.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN; NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Government drawings. The following Government Drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those in effect on the date of the solicitation.

DRAWINGS

ME

TA13220E2823	- Printed Wiring Board Assembly.
TA13220E2850	- Sensor, Capacitance Proximity DT-548/FSS-9 (V).

1/ The "C" revision of MIL-STD-810 shall be used.

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(Copies of drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-JIE, Fort Belvoir, VA 22060-5606.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The capacitance proximity sensor (here-after called "CPS") shall be in accordance with top assembly TA13220E2850 and as specified herein.

3.1.1 Repair parts. The CPS shall contain the following repair parts: printed wiring board (PWB) assembly, signal processor - TA13220E2823. When specified (see 6.2) the PWB assembly shall be furnished separately.

3.1.2 Function. The CPS can detect changes in capacitance of a few picofarads within a few seconds; can accommodate slow changing, humidity-caused capacitance changes, and will alarm if the ground wire is disconnected. The CPS shall detect changes in capacitance to ground of insulated-from-ground metal objects, when the protected objects are approached by a human. The CPS shall detect human bodies or hands approaching within 1 inch of a protected object, and generate an alarm signal when the criteria for detection are met. The CPS shall protect up to 15,000 picofarads (pf) of equipment capacitance (with respect to ground), when the equipment is properly insulated from ground and interconnected with a cable.

3.1.3 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly and operation of the CPS. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

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3.3.1 Material deterioration and control. The CPS shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the CPS may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces, and parts incorporated in the CPS's may be newly fabricated from recovered materials to the maximum extent practicable, provided the CPS produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces, and parts shall not be incorporated in the CPS.

3.4 Environmental conditions. The CPS and PWB assembly when furnished separately shall withstand individual environmental conditions specified herein, or combinations thereof, without damage (see 6.4.2), performance degradation, or malfunction when tested as specified in 4.5.2.7.

3.4.1 Operational.

a. Temperature range of -20 to +150 °F.
b. Humidity up to 95 percent relative.

3.4.2 Nonoperational condition and storage.

a. Temperature range of -30 to +165 °F.
b. Humidity up to 95 percent relative.
c. Shock: bench handling.

3.5 Detailed technical requirements. The CPS and PWB assembly when furnished separately shall perform as specified herein under any of the operating test conditions specified herein.

3.5.1 Terminal board connection. The terminal boards (TB) shall be marked with identifying terminal numbers as follows:

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TB1-	Function
1	Alarm Contact
2	Alarm Contact
3	Spare
4	Spare
5	Tamper switch contact
6	Tamper switch contact
7	+20 Volts direct current
8	20 Vdc return
9	Enclosure ground
10	Spare

On TB1, removable jumpers shall be provided across terminals 2 and 3, 3 and 4, and 4 and 5.

TB2-	Function
1	Sense line input
2	Sense line shield

All terminals shall be protected by the housing and cover to prevent access without removal of the cover.

3.5.2 Tamper protection. The tamper switch shall be assembled into the CPS enclosure such that the switch contacts open with not less than 0.125 inch nor more than 0.250 inch of cover movement and before cover flange is disengaged.

3.5.3 Functional performance

3.5.3.1 Input power. The CPS shall function as specified herein when powered by regulated voltage sources at voltages from 18 (+0, -.2) Vdc to 22 (+.2, -0) Vdc. The CPS shall draw 45 ± 15 milliamps (mA) under all operating conditions. Reverse voltage protection shall be incorporated in the CPS.

3.5.3.2 Amplifier output level. The nominal amplifier output level at TP1 shall be 6.3 ± 0.5 Vdc. The amplifier output shall become stable at this level within 150 seconds after changes in capacitance at the sense line input terminals of up to 5000 picofarads in the high sensitivity position and 15000 picofarads in the low sensitivity position.

3.5.3.3 Alarm output characteristics. An alarm condition shall be signified by a resistance across the alarm contacts of greater than 1 megohm. A non-alarm condition shall be signified by a resistance of less than 10 ohms across these contacts. When the alarm criteria are briefly met, the alarm condition shall exist for not less than 0.5 and not more 5.0 seconds unless input conditions meeting the alarm criteria are still present.

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3.5.3.4 Alarm criteria. The CPS shall display an alarm condition across the alarm contacts when a change in capacitance of 12 ± 1 picofarads in the high sensitivity position and 36 ± 3 picofarads in the low sensitivity position is introduced instantaneously. These criteria shall be met for base capacitances of up 5000 picofarads in high sensitivity and 15,000 picofarads in low sensitivity. The CPS shall not alarm when a change in capacitance of 8 ± 1 picofarads, when in the high sensitivity position, and 18 ± 1 picofarads in the low sensitivity position is introduced instantaneously when the base capacitances are 90 percent or more of their maximum allowable values. The CPS shall also alarm when input power is not present or when the ground wire is disconnected.

3.5.3.5 Security of cables. Security of the cables between the CPS and the protected objects shall be provided by the initiation of a tamper alarm if any part of the sensor cable is cut or shorted. Alarms due to capacitance changes shall not be presented at the tamper terminals. Tamper alarms shall be presented across terminals TB2-5 and TB2-6 and are signified by a resistance of greater than 1 megohm. The not alarm resistance shall be less than 10 ohms. Opening of the tamper switch (see 3.5.2) shall also cause a tamper alarm.

3.6 Burn-in. Each CPS and PWB assembly when furnished separately shall be subjected to a 24 hour burn-in to reduce the number of infant mortality failures (see 4.5.2.4).

3.7 Reliability. The CPS shall demonstrate a mean-time-between-failure (MTBF) of 10,000 hours or greater when tested as specified in 4.5.2.8.

3.8 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

3.9 Identification marking. The CPS and PWB assembly when furnished separately shall be identified in accordance with MIL-STD-130 and the drawings referenced herein.

3.10 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9, except soldering. These requirements are applicable to wiring, welding, brazing, planting, riveting, finishes, machine operations, screw assemblies, and freedom of parts from burrs, sharp edges, or any other damage or defect that could make the part (or equipment) unsuitable for the purposes intended.

3.10.1 Soldering. All soldering shall be performed in accordance with MIL-STD-454, requirement 5.

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3.11 Government-loaned property. Unless otherwise specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.7) for manufacturing and testing the CPS and PWB assembly when furnished separately:

Description	Identification	QTY
Stable base drawing of PWB	13220E2825, sh 2 & 3	1

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.3 Disassembly inspection. Failure of the first article to pass any examination or test shall be cause for disassembly, in the presence of a Government representative, of the first article to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article. Government-loaned property shall be excluded from the requirements of this paragraph.

4.2 Classification of inspection. The CPS and PWB assembly when furnished separately shall be inspected as specified herein. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article CPS and PWB assembly when furnished separately shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection and for performing the disassembly inspection specified in 4.1.3.

4.3.2 Tests.

4.3.2.1 Complete CPS. Complete CPS's shall be subjected to the tests marked "X" in column 1 of table I. Failure of any test shall be cause for rejection and for performing the disassembly inspection specified in 4.1.3.

4.3.2.2 PWB assembly when furnished separately. PWB assemblies when furnished separately shall be subjected to the tests marked "X" in column 1 of table I, except as noted. Failure of any test shall be cause for rejection and for performing the disassembly inspection specified in 4.1.3.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for tests shall be in accordance with MIL-STD-105, inspection level S-2.

4.4.2 Examination. Each CPS and PWB assembly when furnished separately, shall be examined for the major defects specified in 4.5.1. Presence of any defect shall be cause for rejection.

4.4.3 Tests.

4.4.3.1 Individual.

4.4.3.1.1 CPS. Each CPS shall be subjected to the tests marked "X" in column 2 of table I. Failure of any test shall be cause for rejection.

4.4.3.1.2 PWB assembly when furnished separately. Each PWB assembly when furnished separately shall be subjected to the tests marked "X" in column 2 of table I, except as noted. Failure of any test shall be cause for rejection.

4.4.3.2 Samples.

4.4.3.2.1 CPS. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column of 3 of table I. AQL shall be 1.0 percent defective.

4.4.3.2.2 PWB Assembly when furnished separately. Samples selected in accordance with 4.4.1 shall be subjected to the tests marked "X" in column 3 of table I, except as noted. AQL shall be 1.0 percent defective.

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4.5 Inspection procedure.

4.5.1 Examination. The CPS and PWB assembly when furnished separately shall be examined as specified herein for the following defects:

Major

101. Materials not as specified.
102. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes, or treatment.
105. Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the CPS or PWB assemblies when furnished separately.
106. Any dimension, other than those identified on the QAP requirements shown on the drawings, not as specified.
107. Electromagnetic interference suppression components and networks not as specified.
108. Components missing or not as specified.
109. Assembly not correct.
110. Identification or special marking missing, incomplete or illegible.
111. Interchangeability not as specified.
112. Workmanship including wiring and soldering not as specified.
113. Treatment and painting not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified herein, tests shall be conducted at prevailing ambient temperature and humidity conditions. After successful completion of functional testing, each CPS and PWB assembly when furnished separately shall be subjected to a 24-hour burn-in as specified in 4.5.2.4. Unless otherwise specified herein, the CPS and PWB assembly shall operate as specified herein without maintenance.

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4.5.2.2 Test Schedule.

TABLE I. Test schedule.

First Article	Quality Conformance: Individual	Quality Conformance: Sample	Test	Reg Para	Test Para
1	2	3	4	5	6
X	X	-	Burn-in	3.6	4.5.2.4
X	X	-	Functional	3.5.3.1-3.5.3.5	4.5.2.5
X 1/	-	X 1/	Tamper switch tolerance	3.5.2	4.5.2.6
X	-	X	Low Temp	3.4.1(a), 3.4.2(a)	4.5.2.7.1
X	-	X	High Temp	3.4.1(a), 3.4.2(a)	4.5.2.7.2
X	-	X	Humidity	3.4.1(b), 3.4.2(b)	4.5.2.7.3
X	-	-	Shock	3.4.2(c)	4.5.2.7.4
X 1/	-	-	Reliability	3.7	4.5.2.8

1/ Test not required for PWB assemblies when furnished separately.

4.5.2.3 Test methods. Test shall be conducted in accordance with the test methods in the appendix and as specified herein. Calibration of the test equipment shall be in accordance with MIL-STD-45662.

4.5.2.4 Burn-in. Each CPS and PWB assembly when furnished separately shall be subjected to a 24-hour burn-in. During the burn-in, each unit shall be continuously energized in the high sensitivity setting and subjected to the alarm sensitivity tests of appendix test method 1, (4.b) nine times, but no test shall be initiated less than 45 minutes after completion of the previous test period. A complete functional test shall be performed after completion of burn-in. Non compliance with 3.5.3 shall constitute a failure. Failures during burn-in shall be analyzed, repaired, and corrective action taken as necessary. When specified in the contract or purchase order (see 6.2), burn-in test reports shall be forwarded to the contracting officer. Any unit failing during burn-in shall be subjected to an additional 24-hour burn-in after failure analysis and repair and successful completion of the functional test.

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4.5.2.5 Functional. The CPS shall be tested in accordance with appendix test method 1. When furnished separately (see 6.2), the PWB assembly shall be tested in accordance with appendix test method 1.

4.5.2.6 Tamper switch tolerance. The CPS shall be tested in accordance with appendix test method 2. Nonconformance to 3.5.2 shall constitute failure of this test.

4.5.2.7 Environmental. The CPS and PWB assembly when furnished separately shall be subjected to the following series of tests specified in MIL-STD-810. During each test, conduct the functional test specified in 4.5.2.5. At the completion of the environmental tests, conduct the functional tests specified in 4.5.2.5.

4.5.2.7.1 Low temperature. Conduct low temperature test method 502.1, procedure I, storage temperature of -30 °F for not less than 48 hours, equipment operating temperature of -20 °F.

4.5.2.7.2 High temperature. Conduct high temperature test method 501.1, procedure II, except that the storage temperature during step 4 shall be +165 °F and the operating temperature shall be +150 °F.

4.5.2.7.3 Humidity. Conduct test method 507.1, procedure I with the following exceptions:

a. Step 5 shall be for 5 cycles (not less than 120 hours). Perform the functional test at the completion of each cycle.
b. During steps 2 and 3, the relative humidity shall be maintained at 95 percent or greater.
c. During step 4, the relative humidity shall be maintained at 92 percent or greater.
d. Measurement shall commence 1 hour and 15 minutes prior to the completion of each 24 hour cycle. For the tamper test, the covers of the CPS any be wiped of excess external moisture prior to performance of this test. This test may be the last functional test performed prior to continuation of humidity cycling. The CPS may be removed from the environmental chamber for measurement; however, unsuccessful completion of measurements or unsuccessful return of the CPS to the test chamber within 1 hour and 15 minutes subsequent to the initiation of measurements shall constitute failure of the test.

4.5.2.7.4 Shock. Bench handling: Conduct method 516.2, procedure V.

4.5.2.7.5 Failure criteria. Nonconformance to 3.4 shall constitute failure of these tests.

4.5.2.8 Reliability. Using an upper test MTBF 0o of 10,000 hours and a lower test MTBF 01 of 5,000, hours the CPS shall be tested as specified herein, with "accept" and "reject" criteria in accordance with MIL-STD-781, test plan VIIIC.

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Continue testing until either an "accept" or "reject" decision is reached. All test incidents will be scored by the Government. A failure is defined as any malfunction which causes or may cause one or more of the following:

a. Failure to commence operation, cessation of operation, or degradation of performance below the designated levels.
b. Serious damage to the system.
c. Serious personnel safety hazards.

4.5.2.8.1 Test procedure. One or more CPS's shall be randomly selected by a Government representative. The testing shall be conducted under ambient laboratory conditions, without temperature, humidity or vibration cycling. Operation shall be continuous throughout the test with all circuits exercised not less than nine times during each 24-hour period, but not more often than once per hour. This exercising shall consist of performing appendix test method 1. The test shall commence only after successful completion of functional testing and a burn-in period of 24-hours, as specified in 4.5.2.4. Failures chargeable to workmanship deficiencies shall be counted as relevant. Alarms not deliberately excited as part of the equipment test routine shall be reported and cause of the alarm identified, if practicable. This accounting shall be maintained separately from the failure accounting. No adjustments shall be allowed during the test.

4.5.2.8.2 Failure criteria. Nonconformance to 3.7 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of packaging.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

114. Preservation method for complete CPS's and PWB assemblies when furnished separately not as specified for levels A, B and C.
115. Boxed contents not cushioned to prevent movement or damage as specified for level A or B.
116. Consolidation not as specified for level A, B or C.
117. Shipping containers not as specified for level A or B.
118. Quantities packed together in shipping containers exceed gross weight specified for level A, B or C.
119. Strapping not as specified for level A or B.
120. Marking missing, illegible, incorrect, or incomplete for levels A, B, or C.
121. Special marking not as specified.

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5. Packaging.

5.1 Preservation. Preservation shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Each CPS and PWB assembly when furnished separately shall be preserved by one of the following procedures depending upon procurement of complete CPS's or PWB assemblies furnished separately.

5.1.1.1 Procurement of complete CPS's. Each CPS, excluding the RG-58C\U cable and insulating blocks, shall be preserved in accordance with MIL-P-116, method IIc. The cable shall be preserved and the insulating blocks preserved together in accordance with MIL-P-116, method IC-1 or IC-3. The components shall then be placed together in a close-fitting box conforming the PPP-B-636, W6c, style optional. The components shall be cushioned and secured within the box to prevent movement or damage.

5.1.1.2 Procurement of PWB assemblies when furnished separately. Each PWB assembly when furnished separately shall be individually preserved in accordance with MIL-P-116, method IIc, using a close-fitting box conforming to PPP-B-566, or PPP-B-676, variety 1, style, type and class optional.

5.1.1.2.1 Consolidation. PWB assemblies when furnished separately, preserved as specified in 5.1.1.2, shall be consolidated together in a close-fitting box conforming to PPP-B-636, W6c, or W5c, style optional, in quantities not to exceed the weight limitations of the box. The contents shall be cushioned within the box to prevent movement or damage.

5.1.2 Level B. Preservation shall be as specified in 5.1.1 for level A, except method IA-8 shall be used in place of method IIc. Consolidation shall be as specified for level A.

5.1.3 Level C. Each complete CPS and PWB assemblies shall be preserved as specified for level B except the boxes shall be domestic class.

5.2 Packing. Packing shall be level A, level B, or C as specified (see 6.2).

5.2.1 Level A. The CPS's and PWB assemblies when furnished separately, preserved as specified in 5.1, shall be packed by the following procedure depending upon procurement of complete CPS's or procurement of PWB assemblies furnished separately.

5.2.1.1 Packing of complete CPS's. Complete CPS's shall be packed together in a close-fitting box confirming to PPP-B-601, overseas type, grade B, style optional, in quantities not to exceed 200 pounds gross weight. Cushioning and blocking within the box used to prevent movement and isolate impact shock shall be in accordance with MIL-STD-1186. Closure and strapping shall be in accordance with the appendix to the box specification. Strapping shall conform to QQ-S-781, class 1, type I or IV, finish B, size as applicable.

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5.2.1.2 Packing of PWB assemblies when furnished separately. PWB assemblies when furnished separately, preserved as specified in 5.2, shall be packed together as specified in 5.2.1.1.

5.2.2 Level B. Complete CPS's or PWB assemblies when furnished separately, preserved as specified in 5.1 shall be packed as specified in 5.2.1 for level A, except boxes shall be domestic type. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13, or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.2.3 Level C. Complete CPS's or PWB assemblies when furnished separately, preserved as specified in 5.1 shall be packed in boxes conforming to PPP-B-636 domestic class. The quantity of the contents shall not exceed the weight or size limitations of the box. Box closure shall be in accordance with the appendix to the box specification.

5.3 Marking. Marking for military levels of protection shall be in accordance with MIL-STD-129 and the special marking specified in 5.3.1.

5.3.1 Special marking. Each shipping container shall be marked with code labels in accordance with MIL-STD-129. The labels shall contain the project code "JID".

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The CPS is intended for use as a component of the joint-services interior intrusion detection system (J-SIIDS).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2..1.1 and 2.2).
c. Nomenclature and quantity of item(s) to be furnished.
d. Technical Data Package List(s), including revision level, to which the item(s) are to be built.
e. When a first article is required for inspection and approval, the number of units required, and the allocation of units required to environmental and reliability tests respectively (see 3.2).
f. When Government-loaned property is not required (see 3.11).
g. Level of preservation and packing required (see 5.1 and 5.2).
h. Schedule, content, and format of burn-in failure reports (see 4.5.2.1).

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6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions. The following definitions shall apply throughout this specification.

6.4.1 False alarm. An alarm generated by stimuli other than those for which the test item is designed to alarm.

6.4.2 Damage. Damage is defined as deformation, corrosion, loosening of parts, breakage, or change of fit of any components; physical change which impairs the mechanical integrity of the CPS or PWB assembly; evidence of delamination or water penetration into integrated circuits, printed circuit boards, external components, or physical condition resulting in noncompliance of the item with the provisions of this specification.

6.4.3 Alarm signal.

6.4.3.1 Tamper alarm. Alarm resulting from removal of CPS assembly cover or pry-off of the CPS assembly from its mounting.

6.5 Quality assurance provisions (QAP) records. The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of the end items.

6.7 Government-loaned property. The contracting officer should arrange to loan the property specified in 3.11 when applicable (see 6.2).

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6.8 Subject term (key word) listings.

Intrusion detection
Physical security
Sensor

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - ME
Navy - EC
Air Force - 99

Review activity:
Air Force - 82

User activity:
Army - ER

Preparing activity:
Army - ME
Project 6350-0112

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APPENDIX

TEST METHODS

FOR

SENSOR, CAPACITANCE PROXIMITY

10. SCOPE

10.1 Scope. The test methods contained in this appendix are to determine whether CPS's procured under this specification conform to the requirements set forth.

20. APPLICABLE DOCUMENTS. This section is not applicable to this appendix.

WARNING: THE CPS IS VERY SUSCEPTIBLE TO STRAY CAPACITANCE. KEEP LEAD LENGTH SHORT AND WIRE DRESS CONSTANT DURING ALL CALIBRATION AND TESTING PROCEDURES. PERSONNEL NOT INVOLVED IN THE TESTING OF THE CPS SHOULD STAY CLEAR OF THE TEST BED DURING CRITICAL MEASUREMENTS.

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TEST METHOD I

1. Test to be performed.

   a. Input power, amplifier output level, alarm output characteristics, cable security.
   b. Alarm sensitivity.

2. Test apparatus.

   a. Power supply - Hewlett Packard 6217A, or equivalent.
   b. Digital Ammeter (M1) - Hewlett Packard 3466A, or equivalent.
   c. Digital Ohmmeter (2-M2, M3) - Hewlett Packard 3466A, or equivalent.
   d. Digital Voltmeter (M4) - Hewlett Packard 3466A, or equivalent.
   e. Stopwatch
   f. Test fixture (see figure 1).
   g. Test set-up (see figure 2).

3. Test preparation.

   NOTE: Adjustment of amplifier gain is to be performed prior to commencement of Government acceptance tests. No adjustments are allowed once these tests have begun. Gain adjustment is accomplished by setting R29 as necessary to pass the alarm sensitivity test of 4.b. below. This adjustment may require several iterations.

   a. Remove 470K resistor from TB2-1 and TB2-2 on CPS and adjust PS1 to 17.9 ± 0.1 Vdc.
   b. Connect unit under test (UUT) to test equipment as shown in figure 2. Set switches S1 through S8 as follows:

      S1 - Open
      S2 - Hi position (open)
      S3 - Open
      S4 - Open
      S5 - Open
      S6 - Open
      S7 - Open
      S8 - Open

   c. Set sensitivity switch on CPS PWB assembly to high position. When testing PWB assemblies short together the X and Y wires normally connected to the tamper switch.

4. Test Procedure.

   a. Input power, amplifier output level, alarm output characteristics, cable security.

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(1) Turn on power supply PS1. Start timer on stopwatch when PS1 is turned on. Observe M1, M2, M3, and M4. Readings shall be as follows:

M1: 45 ± 15 milliamps (mA).
M2: greater than 1 megohm.
M3: greater than 1 megohm.
M4: varying between greater than 2 and less than 13 volts.

(2) Switch S3 to closed position. Observe M1 and M2. Readings shall be as follows:
M1: 45 ± 15 mA.
M2: less than 10 ohms.

(3) Observe M3 and M4 until M3 becomes less than 10 ohms. Operation shall be as follows:
M3 shall become less than 10 ohms not more than 180 seconds after turning on PS1.
M4 shall become stable at 6.3 ± 0.5 Vdc.

(4) Adjust PS1 to 22.1 ± 0.1 Vdc. Observe M1, M2, M3, and M4. M1 shall be 45 ± 15 milliamps (mA). M2 and M3 shall stabilize to less than 10 ohms. M4 shall stabilize at 6.3 ± .5 V.

(5) Momentarily close S4. M2 shall be greater than 1 megohm while S4 is closed and less than 10 ohms when S4 is open. Disregard M3.

(6) Close S1. Start timer on stopwatch and observe M3. M3 shall be greater than 1 megohm, but shall change to less than 10 ohms in less than 150 seconds.

(7) Momentarily touch terminal TB1-1 on the test fixture. Resistance reading on meter M3 shall be greater than 1 megohm for 0.5 to 5.0 seconds.

b. Alarm Sensitivity Test.

NOTE: An intrusion alarm has occurred if resistance across TB1-1 and TB1-2 (M3) on CPS increases to greater than 1 megohm for at least .5 seconds. A tamper alarm has occurred if the resistance across TB1-5 and TB1-6 (M2) on CPS increases to greater than 1 megohm for at least .5 seconds. When not in alarm, these resistances are less than 10 ohms.

(1) Adjust switches as necessary to obtain following settings:

S1 - Closed
S2 - Hi position
S3 - Closed
S4 - Open
S5 - Open
S6 - Open
S7 - Open
S8 - Open

(2) Adjust PS1 for 20.0 ± 0.2 Vdc.

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(3) Set CPS unit sensitivity switch to HI position.
(4) Allow voltage at TP1 to stabilize at 6.3 ± .5 V (M4).

NOTE: Before proceeding to the next step in the alarm sensitivity test, wait 5 minutes until M4 stabilizes to within ± .2 Vdc of some center value between 5.8 and 6.8 Vdc. Record this center value. Throughout the entire alarm sensitivity test always allow voltage on M4 to stabilize to ± .2 Vdc of this center value before proceeding to the next step in the test.

(5) Switch S5 from Open to Closed position while monitoring M2 and M3. Alarm shall be indicated on M3 only.
(6) After stabilization of M4 switch S5 from Closed back to Open position. An alarm shall occur on M3 only.
(7) Switch S6 from Open to Closed position while observing M2 and M3. No alarms shall occur.
(8) Switch S6 from Closed to Open. No alarms shall occur.
(9) Set CPS sensitivity switch to LO position and test fixture switch S2 to Lo position and allow voltage on M4 to stabilize.
(10) Switch S7 on test fixture from Open to Closed position. An alarm shall occur on M3 only.
(11) Switch S7 from Closed to Open position. An alarm shall occur on M3 only.
(12) Switch S8 from Open to Closed position. No alarms shall occur.
(13) Switch S8 from Closed to Open position. No alarms shall occur.
(14) Momentarily depress S4. M2 shall be greater than 1 megohms while S4 is depressed and less than 10 ohms when S4 is not depressed. Disregard M3.
(15) Switch S3 to Open position, then back to Closed. M2 shall indicate an alarm while S3 is in the Open position and no alarm while S3 is in the Closed position. Disregard M3.
(16) Turn Off power supply and disconnect equipment from test fixture.

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TEST METHOD 2

TAMPER SWITCH TOLERANCE

1. TEST APPARATUS:

a. Power supply - Hewlett Packard 6217A, or equivalent.
b. Digital Ohmmeter (M2) - Hewlett Packard 3466A, or equivalent.

2. TEST PROCEDURE:

Place the CPS in operation by applying 20 Vdc to terminals TB1-7 and TB1-8.

a. Place a 470K ohm resistor across the sensing line (terminals TB2-1 and TB2-2).
b. Connect an ohmmeter between tamper relay contacts TB1-5 and TB1-6.
c. Close the cover of the CPS enclosure. The resistance across the tamper relay contacts shall be less than 10 ohms.
d. The resistance between TB1-5 and TB1-6 shall increase to greater than 1 megohm when the cover is raised more than .125 inch but no more than .250 inch (tamper alarm).
e. The cover flanges shall not be disengaged when the resistance increases to 1 megohm in step 2(d) above.
f. The resistance between TB1-5 and TB1-6 (M2) shall be greater than 1 megohm when the cover is opened (tamper alarm).
g. Pull the interlock switch fully out and lock. The resistance between TB1-5 and TB1-6 shall decrease to less than 10 ohms (no tamper alarm).

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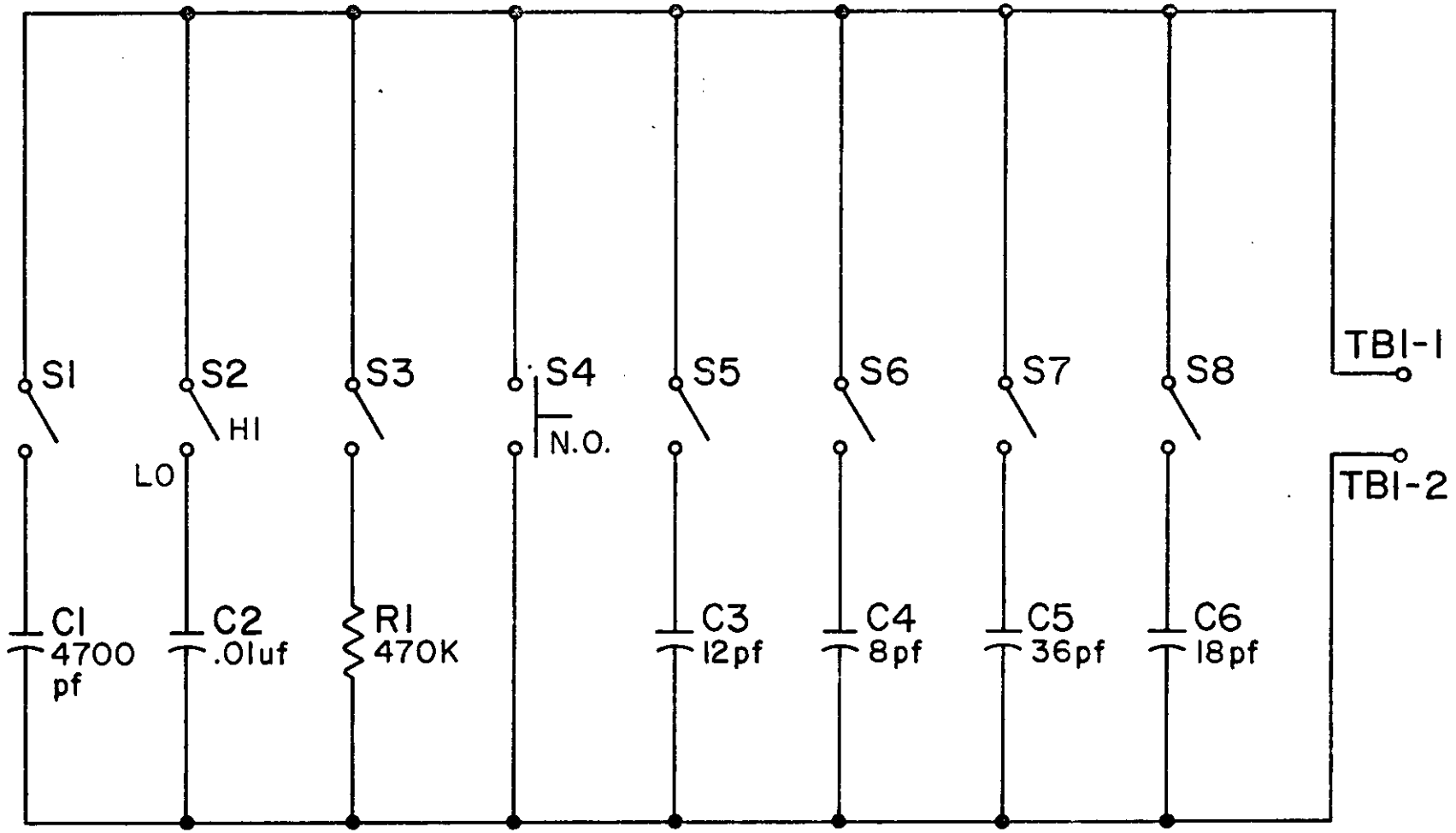


FIGURE 1. Test fixture.

X-4826

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CPS Test Fixture Parts List

C#	Value	Type	Voltage	Tolerance
1	.0047 uf		50 Vdc	±10%
2	.01 uf		50 Vdc	±10%
3	12 pf		50 Vdc	±1 pf
4	8 pf		50 Vdc	±1 pf
5	36 pf		50 Vdc	±3 pf
6	18 pf		50 Vdc	±1 pf

S#	Type
1,2,3,5,6,7,8	SPST Toggle
4	Pushbutton, N.O.

R#	Value (ohms)
1	470 K 1/4 Watt 5%

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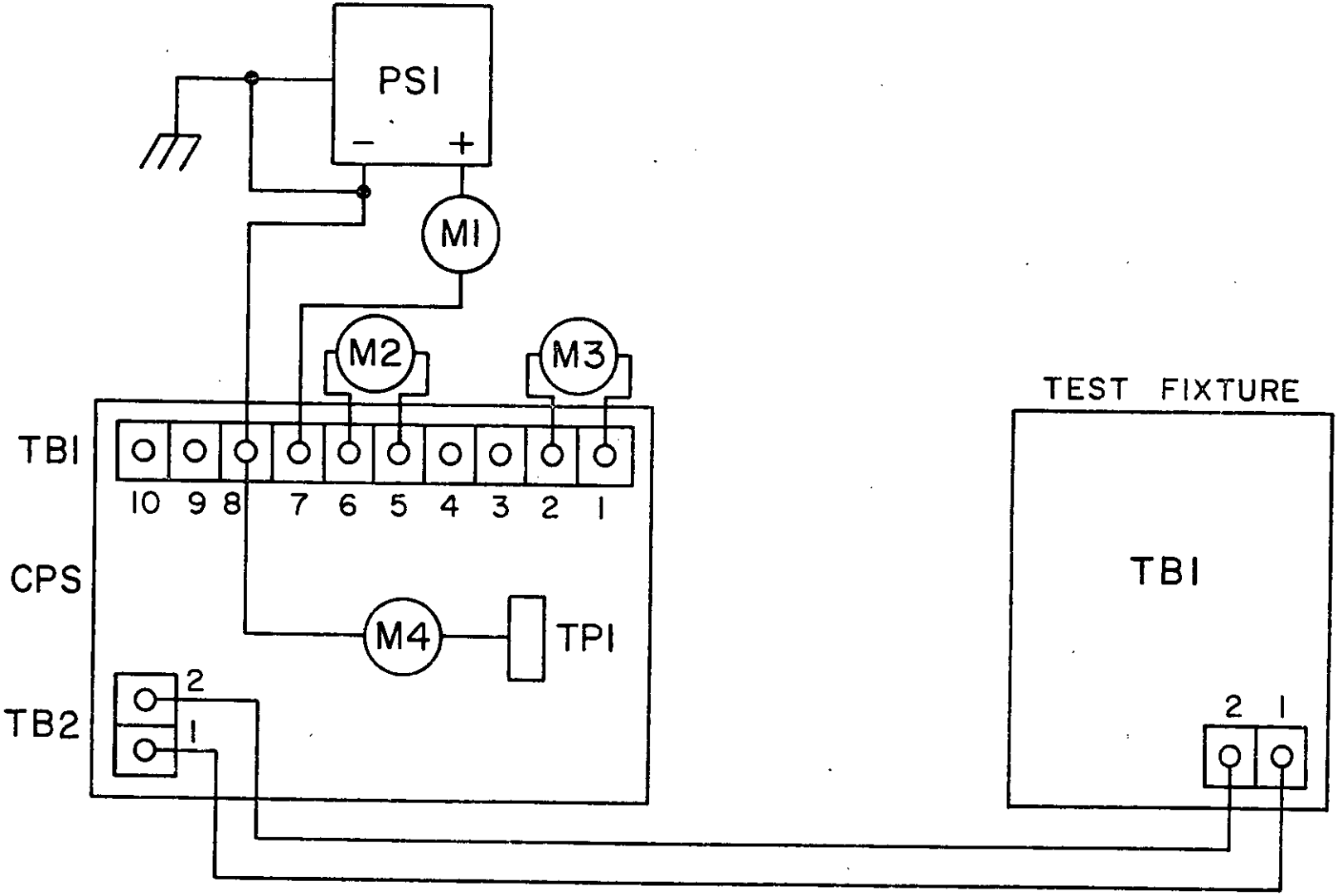


FIGURE 2. Test set-up.

X-4724

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-52868D	Sensor, Capacitance Proximity DT-548/FSS-9(V)

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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